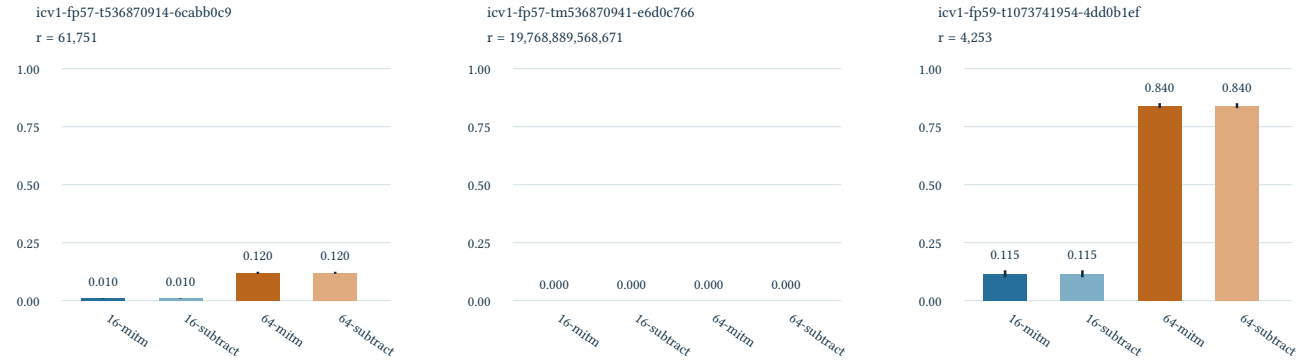
Volcano index calculus: measured control and pilot

Run 20261017-control-pilot. Counts and correctness diagnostic; no runtime or speedup claim.

Direct EC relation yield: audited root-vertex pilot

Two public targets per curve and arm. Mean with observed min–max; no confidence interval.



Measured relation counters; denominator is attempted decompositions. Zero is observed, not a claim of impossibility.

Actual scope: 3 root vertices (conductor 1), 2 public targets each, 8 arms, 1 round, 0 warmups: 48 pilot attempts. Additional small-subgroup control: 12 attempts. This is not graph-wide completion and does not replace the original A–E requirement. A separate resumable survey targets all 24,823 vertices and 51 strata; consult its current ledger rather than this frozen report.

Observed: pilot: 30 verified recoveries, 14 exhausted budgets, 4 watchdog timeouts. Control: 8 verified, 4 exhausted. Both projected-base methods solved both control targets. The raw-abscissa control exhausted its budget in every pilot target. On the actual 45-bit subgroup the projected arms found zero relations in 4,096 trials per target; no smaller subgroup was substituted.

Validation: all four sealed sessions pass the ecbench audit. All 38 successful deterministic records replay identically. Supplemental native replay also reproduces all 18 exhausted records, including every phase/counter, field operation and base identity. Four timeouts have no final counted report; their unknown relation counts are not treated as zero. Every recovered scalar is checked by multiplication in the runner and auditor. The replays use the same implementation and host; independent implementation replay is still open.

Method: cofactor-project each raw abscissa point, reject infinity, check prime-subgroup membership and deduplicate negation pairs. Factor-base sizes 16 and 64; scan cap 20,000; subtraction and two-summand MITM; checked full-rank incremental elimination. Public-target seed 20261017; order seed 1, alternate. Cold base/table setup and verification are charged on every one-target run. References are rho.negation and textbook BSGS, with matched rho A/A. A 3-second child watchdog censors unfinished runs. Both methods use actual largest r.

Counted cold costs and yields

GAE values below are **lower bounds with unpriced native work**. Each row has two measured target attempts; min/mean/max describe those observations only. Exhausted rows are kept separate and do not claim recovered DLP cost.

Source model	Projected base/oracle	Status	n	GAE min	GAE mean	GAE max	Trials sum	Relations sum
icv1-fp57-t536870914-6cabb0c9	16-mitm	verified	2	6,493	6,927.0	7,361	5,400	53
icv1-fp57-t536870914-6cabb0c9	16-subtract	verified	2	76,008	90,295.5	104,583	5,400	53
icv1-fp57-t536870914-6cabb0c9	64-mitm	verified	2	19,530	19,970.0	20,410	2,569	310
icv1-fp57-t536870914-6cabb0c9	64-subtract	verified	2	130,718	160,668.5	190,619	2,569	310
icv1-fp57-tm536870941-e6d0c766	16-mitm	exhausted	2	8,191	8,204.0	8,217	8,192	0
icv1-fp57-tm536870941-e6d0c766	16-subtract	exhausted	2	138,735	138,748.0	138,761	8,192	0
icv1-fp57-tm536870941-e6d0c766	64-mitm	exhausted	2	20,239	20,252.0	20,265	8,192	0
icv1-fp57-tm536870941-e6d0c766	64-subtract	exhausted	2	536,271	536,284.0	536,297	8,192	0
icv1-fp59-t1073741954-4dd0b1ef	16-mitm	verified	2	4,242	4,419.0	4,596	598	67
icv1-fp59-t1073741954-4dd0b1ef	16-subtract	verified	2	10,148	12,328.0	14,508	598	67
icv1-fp59-t1073741954-4dd0b1ef	64-mitm	verified	2	20,463	20,467.5	20,472	532	447
icv1-fp59-t1073741954-4dd0b1ef	64-subtract	verified	2	22,087	22,588.0	23,089	532	447

Complete accounting: native summary.json and statistics.csv include min, max, arithmetic mean and explicit n by graph, conductor, arm, status and phase; observations.jsonl retains record/workload ids and all phase-native counters. Raw sessions retain every field operation, phase, setup, matrix rank/dependency, target window and failure. pricing-band.json applies common 0/0.1/1 GAE native-work weights to IC and reference arms alike. No calibrated versus uncalibrated speed ranking is inferred here.

Host and limits: Intel Xeon Platinum 8573C, Linux x86-64 container, four-CPU quota and 16 GiB RAM. ecbench reserves CPU 4; every measured record earns L0: unoptimized build, movable-thread/PSI and host-isolation limitations. Raw in-process and end-to-end elapsed observations are retained. None is an admitted optimized runtime distribution. Two targets/one round also do not qualify wall-clock comparisons. PMU instructions/cycles were unavailable.

Provenance: projected-base source/tests and first runner commit 4eecf6b53; exhaustion replay and survey ordering commit ee54ad603. Exact ecbench SHA-256: 1869ca2e7be271794df8128c08a6dca0201038e2c81a3e0ce6434f7ec0d8379d. Per-campaign manifests bind source, binary, queue and input hashes. Exact control/pilot protocols, frozen specs, plans, records and replay receipts accompany this report. Compiler/test exit logs are retained. New subgroup base tests pass 2/2, including every nonzero control logarithm without a planted-answer input. Conductor reservation was unavailable (exit 127).

Open requirements: exact per-vertex representation registry, optimized and qualified full-matrix execution, all targets/rounds/base sizes, transported edge experiments, composite-order PH, independent implementation replay and finite-field IC. Actual embedding degrees are 30,875; 3,953,777,913,734; and 2,126. The existing supersingular degree-2 teaching MOV backend cannot execute finite-field IC in those destinations. This is an implementation gap, not proof that mathematical reductions do not exist. Original A/C/D/E full generation remains storage-blocked; no original case is replaced here.

No new mathematical theorem or fastest-method claim is made. The established IC/rho frontier remains unchanged. These results establish the measured control/pilot behavior, not unrun strata or vertices.